

Name:

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Section:

STOICHIOMETRY OF THE REACTION WITH STRONG ACID LAB WORK-SHEET

Show all work (except average and standard deviation). Turn in the post-lab assignment before leaving the lab. Reminder: never round values prematurely; only round to the correct number of significant figures when reporting final answers and use unrounded values for subsequent calculations.

Prepare an Excel spreadsheet to tabulate and compute the answers to the following questions; for each question, you will need to show all work for **just 1 trial** - remember that doing the first calculation by hand will make it easier to write the spreadsheet formula for the remaining trials.

1. Data analysis

- (a) Show a full sample calculation for the moles of $\text{H}_2(g)$ collected for one trial. (Remember to calculate the rest in your spreadsheet).

$n_{\text{H}_2}(\text{mol}) =$

- (b) Compute the stoichiometric factor (x) for the same trial – the ratio of H reacting (twice the number of moles of H_2 generated) to the number of moles of Mg reacting.

$x =$

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- (c) Report the average stoichiometric ratio and standard deviation. Compute the relative standard deviation in the stoichiometric factor.

$\bar{x} =$

RSD =

- (d) Can any of the trials be statistically rejected using a Grubbs test? If so, reject the point(s) and recalculate the mean and standard deviation. Show work for the Grubbs analysis.

Reject a trial? (yes/no)

Recalculated $\bar{x} =$

- (e) Using your *best* average (after the Grubbs test), calculate and report the 95% confidence interval for the stoichiometric ratio, showing your work.

$\bar{x} \pm \Delta =$

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- (f) Report the balanced equation for magnesium with hydronium ion using your experimental value for the stoichiometric ratio. Do not round your value – use the appropriate number of experimental significant figures.

2. Statistical analysis

- (a) Compute the percent relative difference, using the appropriate formula, between your value of the stoichiometric ratio to the expected value. Explain your choice of formula in no more than 1 sentence.

%diff =

- (b) According to your percent relative difference, is your observed stoichiometric ratio consistent with the expected value? Explain in no more than 1 sentence.
- (c) Compare your value of the stoichiometric ratio to the expected value using your confidence interval. Is your value statistically consistent with the theoretical value? Explain in no more than 1 sentence.

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- (d) Which is the more appropriate measure of accuracy to use in this experiment: confidence interval or percent relative difference? Explain your choice.

3. Questions for thought

- (a) Why was it important to know the barometric pressure in the lab and the ambient temperature? What would have occurred if you hadn't recorded this information?
- (b) "Throwing out" data is tricky. Sometimes it is okay to just ignore a data point recorded, and other times you are required to perform a statistical analysis in order to determine if a point is to be discarded. Provide an example of each of these two scenarios, where data is *properly* discarded in these manners. What is the major difference between these two methods of discarding data? (Hint: consider the reason why the data is being thrown out.)